

THOMSON, GA, USA

WMO: 720289

Lat: 33.530N

Lon: 82.516W

Elev: 152

StdP: 99.51

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63836

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-2.9	-15.7	1.0	3.0	-12.4	1.3	3.4	8.7	8.7	7.8	8.3	0.7	300	0.359

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	36.0	23.6	34.5	23.7	33.3	23.7	26.8	31.5	26.1	30.8	25.5	30.2	2.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.8	21.5	28.7	25.0	20.5	28.1	24.3	19.5	27.4	84.8	31.7	81.8	30.5	79.1	30.0	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	5.7	5.0	-8.7	37.2	2.3	1.9	-10.3	38.5	-11.6	39.6	-12.9	40.6	-14.5	42.0		
(5)			-8.9	27.3	2.3	1.6	-10.6	28.4	-12.0	29.4	-13.2	30.3	-14.9	31.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	7.6	9.8	13.1	17.5	21.7	25.8	27.1	26.5	23.9	18.0	11.6	9.2	(6)	
	DBStd	8.06	5.52	5.00	5.03	4.22	3.43	2.56	2.07	2.21	2.89	4.33	4.50	5.07	(7)	
	HDD10.0	316	114	62	26	3	0	0	0	0	0	2	32	78	(8)	
	HDD18.3	1379	334	243	174	65	12	0	0	0	2	60	206	284	(9)	
	CDD10.0	3124	39	54	123	227	364	473	530	512	417	248	81	54	(10)	
	CDD18.3	1147	2	2	13	39	117	223	272	253	169	48	5	3	(11)	
	CDH23.3	11098	4	17	134	434	1158	2270	2772	2348	1495	421	35	10	(12)	
	CDH26.7	4826	0	1	23	109	447	1094	1347	1062	634	107	2	0	(13)	
(14) Wind	WSAvg	1.7	2.0	2.1	2.1	2.1	1.8	1.5	1.4	1.3	1.4	1.5	1.6	1.8	(14)	
(15) Precipitation	PrecAvg	1145	111	104	126	69	78	107	110	113	94	80	74	84	(15)	
	PrecMax	1414	200	176	238	178	221	201	218	190	288	253	220	168	(16)	
	PrecMin	892	50	43	29	22	9	37	32	37	9	3	18	24	(17)	
	PrecStd	165	41	36	56	39	50	53	52	57	67	63	49	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.8	25.9	28.9	31.3	35.1	37.3	37.4	37.3	35.3	31.9	26.5	24.8	(19)	
		MCWB	17.8	18.3	17.8	19.1	21.3	22.4	23.5	22.7	22.9	23.3	18.9	19.8	(20)	
	2%	DB	21.1	22.9	27.1	29.5	32.7	35.5	36.0	34.9	33.4	29.3	24.0	22.6	(21)	
		MCWB	16.7	15.9	17.4	18.8	21.4	23.6	24.4	23.7	22.9	21.1	18.0	18.9	(22)	
	5%	DB	18.9	21.0	24.5	27.7	31.1	33.9	34.1	33.6	32.3	27.6	22.4	20.3	(23)	
		MCWB	15.7	14.8	16.0	17.7	21.2	23.2	24.1	24.5	22.7	20.8	17.0	17.9	(24)	
	10%	DB	17.1	18.3	22.4	26.1	29.0	32.4	32.9	32.3	30.7	25.9	20.3	18.1	(25)	
		MCWB	13.9	13.8	15.4	17.2	20.8	23.2	24.3	24.0	22.6	19.5	16.5	15.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	19.8	20.7	22.6	24.7	26.8	27.7	27.7	27.2	25.4	22.6	22.2	(27)	
		MCDB	22.2	23.4	25.3	27.3	29.8	32.9	33.3	32.4	30.1	27.3	23.9	23.3	(28)	
	2%	WB	17.8	18.4	19.2	21.3	23.6	25.9	26.8	26.6	25.6	23.7	20.5	20.1	(29)	
		MCDB	19.8	21.5	23.3	25.4	28.8	31.8	31.7	31.0	29.6	26.8	22.3	21.0	(30)	
	5%	WB	16.4	16.9	18.0	20.1	22.7	25.3	26.1	25.8	24.6	22.8	18.9	18.3	(31)	
		MCDB	18.2	19.4	22.2	24.2	27.8	30.9	30.7	30.1	28.7	25.6	21.0	19.9	(32)	
	10%	WB	14.6	15.3	16.8	19.0	21.9	24.6	25.5	25.2	23.8	21.4	17.0	16.3	(33)	
		MCDB	16.6	17.4	20.9	23.2	26.5	29.6	30.3	29.5	27.9	24.2	19.4	18.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	12.3	13.5	13.8	12.8	11.9	11.2	10.7	11.8	13.1	13.5	11.7	(35)	
		MCDBR	13.2	14.1	15.3	15.3	14.8	14.1	12.9	12.7	14.0	14.1	15.5	13.0	(36)	
	5% WB	MCWBR	9.0	8.3	7.3	6.1	5.5	4.3	4.1	4.3	4.9	7.1	9.4	9.2	(37)	
		MCWBR	10.7	11.0	12.3	11.8	11.9	11.6	11.0	10.6	11.1	10.5	11.2	10.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.318	0.325	0.360	0.393	0.455	0.489	0.532	0.525	0.429	0.390	0.345	0.315	(40)	
		taud	2.513	2.490	2.421	2.313	2.184	2.136	2.036	2.042	2.296	2.379	2.460	2.551	(41)	
	Ebn at Noon		899	928	916	895	839	807	770	768	834	842	855	882	(42)	
		Edh at Noon	84	95	111	129	148	155	171	167	124	106	88	77	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.80	3.40	4.50	5.62	6.18	6.35	6.05	5.53	4.82	4.05	3.07	2.41	(44)	
	RadStd		0.15	0.38	0.33	0.38	0.50	0.47	0.43	0.26	0.43	0.48	0.26	0.24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+184	+130

Nomenclature: See separate page